

MRC Lecture 2 — Derivation Guide 3

The $\sqrt{d_k}$ Scaling Argument: Why Temperature Is Necessary

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1. Motivation

In Derivation Guide 2, we derived the attention equation and divided the score matrix by $\sqrt{d_k}$ before applying softmax. This scaling factor was introduced by Vaswani et al. (2017) with a brief statistical justification. In this guide, we derive the argument fully: we prove that the variance of the dot product grows linearly with d_k , show that this causes softmax to saturate without scaling, and demonstrate that dividing by $\sqrt{d_k}$ restores unit variance.

The result connects directly to the temperature parameter from Lecture 1 (Derivation Guide 5): the scaling factor $\sqrt{d_k}$ IS the temperature τ , controlling the sharpness of selection. Vaswani chose it for numerical stability. The Boltzmann distribution uses $k_B T$ for the same mathematical role.

Understanding why scaling is necessary is understanding why temperature exists.

2. Setup: Random Queries and Keys

Assumption. Consider a single query vector $q = (q_1, q_2, \dots, q_{d_k})$ and a single key vector $k = (k_1, k_2, \dots, k_{d_k})$, where each component is an independent random variable with:

$$\begin{aligned} E[q_l] &= 0, \text{Var}(q_l) = 1 \text{ for all } l \\ E[k_l] &= 0, \text{Var}(k_l) = 1 \text{ for all } l \end{aligned}$$

and all q_l and k_l are mutually independent.

Note. These assumptions are approximately satisfied after standard initialization schemes (Xavier/Glorot initialization, which sets weights so that activations have zero mean and unit variance) and layer normalization (which normalizes activations to zero mean and unit variance within each layer). They are modeling assumptions for the purpose of this derivation, not exact properties of trained networks.

3. The Mean of the Dot Product

Claim. $E[q \cdot k] = 0$.

Proof. The dot product is:

$$q \cdot k = \sum_{l=1}^{d_k} q_l k_l$$

By linearity of expectation:

$$E[q \cdot k] = \sum_{l=1}^{d_k} E[q_l k_l]$$

Since q_l and k_l are independent:

$$E[q_l k_l] = E[q_l] \cdot E[k_l] = 0 \cdot 0 = 0$$

Therefore:

$$E[q \cdot k] = \sum_{l=1}^{d_k} 0 = 0 \quad \blacksquare$$

The expected dot product between a random query and a random key is zero. On average, random vectors are orthogonal — neither compatible nor incompatible.

4. The Variance of the Dot Product

Claim. $\text{Var}(q \cdot k) = d_k$.

Proof. The variance of a sum of independent random variables is the sum of their variances:

$$\text{Var}(q \cdot k) = \text{Var}\left(\sum_{l=1}^{d_k} q_l k_l\right) = \sum_{l=1}^{d_k} \text{Var}(q_l k_l)$$

We need $\text{Var}(q_l k_l)$. Using the identity $\text{Var}(X) = E[X^2] - (E[X])^2$:

$$\text{Var}(q_l k_l) = E[(q_l k_l)^2] - (E[q_l k_l])^2 = E[q_l^2 k_l^2] - 0^2 = E[q_l^2 k_l^2]$$

Since q_l and k_l are independent:

$$E[q_l^2 k_l^2] = E[q_l^2] \cdot E[k_l^2]$$

Now, $E[q_l^2] = \text{Var}(q_l) + (E[q_l])^2 = 1 + 0 = 1$, and similarly $E[k_l^2] = 1$.

Therefore:

$$\text{Var}(q_l k_l) = 1 \cdot 1 = 1$$

Summing over all d_k components:

$$\text{Var}(q \cdot k) = \sum_{l=1}^{d_k} 1 = d_k \blacksquare$$

The variance of the dot product is exactly d_k . The standard deviation is $\sqrt{d_k}$.

5. Consequences: The Saturation Problem

The magnitude of typical dot products. Since $q \cdot k$ has mean 0 and standard deviation $\sqrt{d_k}$, typical dot product values are of order $\pm\sqrt{d_k}$.

Numerical examples:

Key dimension d_k	Std. dev. $\sqrt{d_k}$	Typical dot products
4	2	± 2
64	8	± 8
128	11.3	± 11
512	22.6	± 23

Effect on softmax. Recall from Lecture 1 (Derivation Guide 4, §11) that the probability ratio between two candidates is:

$$\frac{P_A}{P_B} = \exp(z_A - z_B)$$

If typical scores are ± 23 (at $d_k = 512$), then a typical score difference is of order 46. The probability ratio is:

$$\exp(46) \approx 9.5 \times 10^{19}$$

This means the highest-scoring key receives essentially all the probability ($P \approx 1$), and all other keys receive $P \approx 0$. The softmax output is a near-one-hot vector.

Why this is a problem:

1. **Loss of information.** The attention output becomes approximately equal to a single value vector ($o_i \approx v_{\text{argmax}}$) rather than a weighted average. The model cannot blend information from multiple positions.

2. **Vanishing gradients.** When softmax saturates, the gradient $\partial P_j / \partial z_j$ approaches zero for all j (both the winner and the losers). The model stops learning because gradient descent has no signal to follow.
3. **Effective argmax.** The soft, probabilistic selection that makes attention powerful degenerates into hard, deterministic selection. The model loses the ability to express uncertainty about which positions are relevant.

6. The Solution: Scale by $\sqrt{d_k}$

Claim. Dividing the dot product by $\sqrt{d_k}$ restores unit variance regardless of d_k .

Proof.

$$\text{Var}\left(\frac{q \cdot k}{\sqrt{d_k}}\right) = \frac{1}{d_k} \text{Var}(q \cdot k) = \frac{d_k}{d_k} = 1 \quad \blacksquare$$

After scaling, the typical score values are of order ± 1 , regardless of the key dimension. The softmax operates in its sensitive regime where score differences produce graded (not saturated) probabilities.

Numerical verification at $d_k = 512$:

	Without scaling	With scaling
Typical scores	± 23	± 1
Typical score difference	~ 46	~ 2
Probability ratio	$\exp(46) \approx 10^{20}$	$\exp(2) \approx 7.4$
Softmax output	Near one-hot	Graded distribution
Gradients	Vanishing	Meaningful

7. Scaling as Temperature

The scaled dot product can be written as softmax with temperature $\tau = \sqrt{d_k}$:

$$A_{ij} = \frac{\exp(q_i \cdot k_j / \sqrt{d_k})}{\sum_k \exp(q_i \cdot k_k / \sqrt{d_k})} = \text{softmax}(q_i \cdot k_j / \tau) \text{ with } \tau = \sqrt{d_k}$$

This is exactly the temperature-scaled softmax from Lecture 1 (Derivation Guide 5). The three regimes:

- **Small $d_k \rightarrow$ small $\tau \rightarrow$ sharp attention.** With $d_k = 4$ ($\tau = 2$), the attention is relatively sharp — the model concentrates on a few positions.
- **Large $d_k \rightarrow$ large $\tau \rightarrow$ diffuse attention.** With $d_k = 512$ ($\tau \approx 23$), the attention is relatively diffuse — many positions receive non-negligible weight.
- **The scaling normalizes this.** After dividing by $\sqrt{d_k}$, the effective sharpness is independent of d_k . The temperature $\tau = \sqrt{d_k}$ exactly compensates for the growth of the dot product variance.

The Boltzmann parallel. In statistical mechanics, the temperature $k_B T$ plays the same role. At low temperature, the Boltzmann distribution concentrates on the lowest-energy state (sharp selection). At high temperature, it distributes probability more uniformly (diffuse selection). The temperature prevents the system from locking into its ground state and allows thermal exploration of the energy landscape. The $\sqrt{d_k}$ scaling prevents the attention mechanism from locking into a single key and allows graded attention across all positions. The mathematical role is identical.

8. Worked Example: Scaling at Different Dimensions

Problem. Two tokens have query-key dot products that happen to differ by the same “true compatibility difference” regardless of dimension. Specifically, token A has dot product $1.5\sqrt{d_k}$ with

a key, and token B has dot product $0.5\sqrt{d_k}$ with the same key (the “true” score difference is $1.0\sqrt{d_k}$, scaling with dimension as expected for random vectors). Compare the softmax output with and without $\sqrt{d_k}$ scaling at $d_k=4$, $d_k=64$, and $d_k=512$.

Without scaling (raw dot products as softmax input):

d_k	Score A	Score B	Difference e	$P_A / P_B = \exp(\Delta)$
4	3.0	1.0	2.0	7.4
64	12.0	4.0	8.0	2,981
512	33.9	11.3	22.6	6.5×10^9

Without scaling, the probability ratio explodes as d_k increases. At $d_k=512$, token A receives 6.5 billion times more probability than token B — effectively a one-hot distribution.

With scaling (dot products divided by $\sqrt{d_k}$):

d_k	Scaled A	Scaled B	Difference	$P_A / P_B = \exp(\Delta)$
4	1.5	0.5	1.0	2.72
64	1.5	0.5	1.0	2.72
512	1.5	0.5	1.0	2.72

After scaling, the probability ratio is 2.72 regardless of d_k . The scaling perfectly compensates for the growth of dot product magnitudes with dimension. The attention distribution maintains the same sharpness at every dimension.

9. Summary

Concept	Result	Significance
Mean of $q \cdot k$	$E[q \cdot k] = 0$	Random vectors are orthogonal on average
Variance of $q \cdot k$	$\text{Var}(q \cdot k) = d_k$	Dot product magnitude grows with dimension
Std. dev. of $q \cdot k$	$\text{SD} = \sqrt{d_k}$	Typical scores are $\pm \sqrt{d_k}$
Without scaling	Softmax saturates at large d_k	Near-one-hot output, vanishing gradients
After scaling by $\sqrt{d_k}$	$\text{Var}(q \cdot k / \sqrt{d_k}) = 1$	Unit variance regardless of d_k
Scaling = temperature	$\tau = \sqrt{d_k}$	Same mathematical role as $k_B T$

10. Checkpoint Questions

Q1. If the query and key components have variance σ^2 instead of 1 (e.g., after a different initialization), what scaling factor should be used?

Answer: $\text{Var}(q \cdot k) = d_k \sigma^4$ (since $\text{Var}(q_l k_l) = E[q_l^2] E[k_l^2] = \sigma^2 \cdot \sigma^2 = \sigma^4$). The standard deviation is $\sigma^2 \sqrt{d_k}$. To restore unit variance, divide by $\sigma^2 \sqrt{d_k}$. If layer normalization ensures $\sigma = 1$, this reduces to $\sqrt{d_k}$ as derived.

Q2. A transformer is designed with $d_k=256$. What is the temperature τ ? If the designer switches to $d_k=1024$ without changing the scaling, what happens to the attention distributions?

Answer: At $d_k=256$: $\tau = \sqrt{256} = 16$. At $d_k=1024$: $\tau = \sqrt{1024} = 32$. If the designer keeps scaling by $\sqrt{d_k}$, the attention distributions maintain the same sharpness. If the designer forgets to update the scaling (still divides by 16 instead of 32), the effective temperature is too low — scores are divided by 16 but their variance is now 1024, so the scaled scores have variance $1024/256=4$ instead of 1. The attention distributions would be sharper than intended.

Q3. In the SFM framework, the temperature is a learned parameter τ_{learned} rather than fixed at $\sqrt{d_k}$. What advantage does this provide?

Answer: A learned τ can adapt to the actual score distribution, which may differ from the theoretical prediction (e.g., after training, the query and key distributions may no longer be zero-mean with unit variance). More importantly, different molecular recognition systems may require different selection stringencies — CRISPR needs very sharp selection while innate immune receptors need broader selection. A fixed $\sqrt{d_k}$ imposes the same sharpness on all systems. A learned τ allows each SFM to discover the selection stringency appropriate to its domain. The temperature recovery prediction (Lecture 10) tests whether τ_{learned} has the physical meaning of $k_B T$.

11. Connection to Future Lectures

- **Lecture 4:** The Boltzmann distribution is derived from maximum entropy with $\tau = k_B T$. The physical temperature controls selection stringency for the same mathematical reason that $\sqrt{d_k}$ controls attention sharpness: without temperature, the system locks into its lowest-energy state.
- **Lecture 6:** The convergence equation $P(j) = \exp(s_j / \tau) / \sum \exp(s_k / \tau)$ is proven unique. Temperature τ is the only free parameter — the functional form is fully determined by five axioms.
- **Lecture 7:** SFMs use learned temperature τ following CLIP. The value of τ_{learned} is a prediction of the theory, testable via affinity calibration.
- **Lecture 10:** The temperature recovery prediction $w_B / w_A = \tau_{\text{learned}}$ provides a quantitative test of whether the learned temperature has physical meaning.